

An Explainable AI Model for IoMT Network Attack Detection

Ralph Lickley

Student ID: 730030331

Abstract

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☐ I certify that all material in this dissertation which is not my own has been identified.

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1 Introduction

With the rapid expansion of the Internet of Things (IoT), many of our devices have become increasingly interconnected, where the range of device types now connected to each other has largely expanded. As a consequence of this, the Internet of Medical Things (IoMT) has inspired significant development in the use of this technology in healthcare settings. This ranges from wearable fitness tracking devices, heart rate monitoring among many other applications.

2 Use of AI tools in AI-Assisted Assessments

For AI-Assisted assessments, you may use GenAI tools if you choose to do so. This is because using AI tools in specific ways can assist with your learning and will not inhibit fair assessment of your achievement of the module's intended learning outcomes [2].

This means: You may use GenAI tools ethically and responsibly in accordance with the purposes in the checked boxes in your assessment brief.

When writing your assessment, you must never use AI tools:

1. For uses other than those represented by checked boxes in your assessment brief.
2. To translate more than a word or short phrase into English unless agreed in the checked boxes in your assessment brief.
3. To upload sensitive or identifying material to an AI tool.
4. To present material that has been generated by AI as your own work or the work of someone else.

When submitting your assessment, you must:

1. Check the box during the submission process, that confirms you have adhered to the university's academic conduct policy and the expectations on use of GenAI in your assessment brief.
2. Treat the AI tool like a citation from any other source.
3. Include a list of all AI prompts and, where possible, hyperlinks to their output with your references, at the end of your work. You do not need to include the outputs themselves, just the links.
4. Retain the full outputs generated by the prompts you have used during your assignment for your records. These outputs should be accessible through the hyperlinks which you have submitted with your assignment. If hyperlinks are not produced through the tools you use, save the AI-generated output in a document that you can easily access later (e.g. on OneDrive). You may be asked to produce this material in the event of an academic conduct inquiry.

Here is a template [3] that you may use to record your AI use when completing AI-integrated and AI-assisted assignments. You can also create your own document to record AI use if you prefer.

2.1 How to reference GenAI using a specific referencing style

Find your referencing style and follow the links to Cite Them Right (CTR) [4]. Once you are on the Cite Them Right homepage for your referencing style, go to the 'Digital and Internet' tab. You should see a link for Generative AI. Sometimes this is listed under 'Software'. If your department is using a referencing style that is not included in Cite them right, you should visit the guidance for your style and ask your module convenor for further advice.

3 An example section

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Table 1: Example of a table.

Country List			
Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO numeric Code
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

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3.1 An example sub-section

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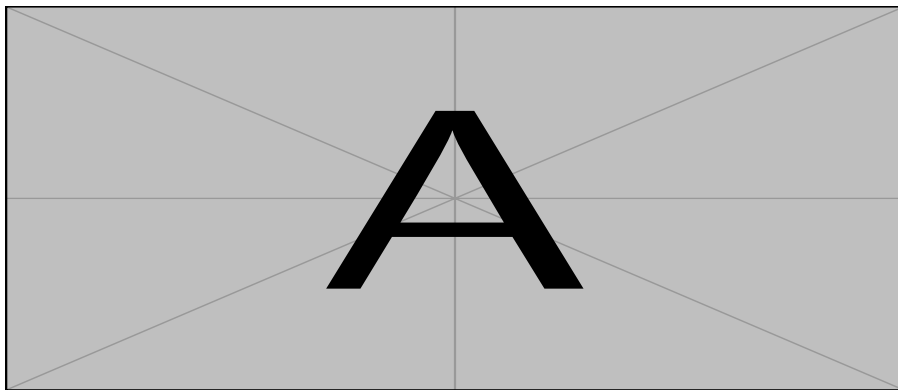


Figure 1: Example of a figure.

References

- [1] “Assessments,” University of Exeter. Accessed: September 23, 2025. [Online]. Available: <https://www.exeter.ac.uk/students/infopoints/yourinfopointservices/assessments/>
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- [5] S. Brin and L. Page, “The anatomy of a large-scale hypertextual web search engine,” in *Computer networks and ISDN systems*, vol. 30, no. 1–7, pp. 107–117, 1998.
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